

DEM'YANSKOYE, Russian Federation

WMO: 280760

Lat: 59.5921N

Lon: 69.3445E

Elev: 57

StdP: 100.64

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.8	(c) -33.9	(d) -39.8	(e) 0.1	(f) -36.6	(g) -36.7	(h) 0.1	(i) -33.7	(j) 5.1	(k) -21.4	(l) 4.4	(m) -20.3	(n) 0.8	(o) 80	(p) 0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.7	(c) 28.4	(d) 20.0	(e) 26.8	(f) 19.3	(g) 25.0	(h) 18.4	(i) 21.3	(j) 26.4	(k) 20.2	(l) 24.9	(m) 19.2	(n) 23.6	(o) 1.9	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.6	(b) 14.4	(c) 23.9	(d) 18.6	(e) 13.5	(f) 22.7	(g) 17.5	(h) 12.6	(i) 21.6	(j) 62.3	(k) 26.4	(l) 58.5	(m) 25.0	(n) 55.2	(o) 23.7	(p) 25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.3	4.4	3.9	-38.1	31.5	4.0	1.0	-41.0	32.2	-43.4	32.8	-45.7	33.3	-48.6	34.0
			-38.4	23.1	4.0	0.9	-41.3	23.8	-43.6	24.4	-45.8	24.9	-48.7	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.2	-19.3	-16.4	-6.8	0.9	8.6	15.6	17.6	14.4	8.8	1.6	-8.7	-15.3		
	DBStd	14.09	8.54	7.92	6.25	5.82	5.65	4.85	3.93	4.05	4.16	5.23	7.78	8.57		
	HDD10.0	4237	907	740	521	276	97	9	1	7	71	262	562	784		
	HDD18.3	6712	1166	973	779	524	304	109	63	134	286	518	812	1042		
	CDD10.0	651	0	0	0	2	53	176	237	143	36	3	0	0		
	CDD18.3	85	0	0	0	0	3	27	41	13	1	0	0	0		
	CDH23.3	749	0	0	0	0	53	248	353	92	3	0	0	0		
	CDH26.7	152	0	0	0	0	11	52	75	14	0	0	0	0		
(14) Wind	WSAvg	1.7	1.3	1.4	1.7	2.0	2.1	1.9	1.7	1.6	1.6	1.7	1.6	1.4		
(15) Precipitation	PrecAvg	584	28	24	28	29	47	67	77	90	65	47	44	38		
	PrecMax	822	52	46	71	65	89	118	275	154	129	124	71	71		
	PrecMin	417	1	2	0	2	8	31	23	35	13	3	6	16		
	PrecStd	88	14	10	15	14	21	26	49	37	27	26	17	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.1	1.6	7.7	18.5	28.2	30.4	30.5	28.4	23.1	16.6	5.7	1.2		
		MCWB	-0.3	0.4	3.3	11.3	16.2	20.7	21.3	21.0	16.9	11.1	4.6	0.7		
	2%	DB	-2.7	-0.4	4.6	14.0	24.7	28.2	28.6	25.9	19.7	13.4	3.3	-0.4		
		MCWB	-3.1	-1.3	1.1	8.7	15.2	19.6	20.5	19.7	15.1	9.6	2.7	-0.8		
	5%	DB	-5.2	-3.3	2.6	11.5	21.9	26.4	27.1	23.6	17.3	10.8	1.6	-2.0		
		MCWB	-5.7	-4.3	0.0	6.9	13.7	18.8	19.9	18.6	13.6	8.4	1.1	-2.4		
	10%	DB	-7.3	-5.6	1.1	9.1	18.7	24.2	25.2	21.4	15.1	8.7	0.3	-4.3		
		MCWB	-7.7	-6.5	-0.7	5.3	12.2	17.9	19.0	17.3	12.4	7.0	-0.3	-4.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	0.9	3.7	11.8	17.8	22.2	23.0	22.1	17.7	11.8	4.7	0.9		
		MCDB	0.1	1.6	7.3	16.7	25.2	28.5	28.4	27.3	21.2	15.1	5.6	1.2		
	2%	WB	-3.1	-1.3	1.6	9.6	15.8	20.6	21.8	20.4	15.6	10.1	2.8	-0.6		
		MCDB	-2.7	-0.3	3.8	13.5	23.2	26.1	26.9	24.5	18.6	12.9	3.3	-0.3		
	5%	WB	-5.5	-4.2	0.5	7.3	14.3	19.5	20.7	19.2	14.1	8.6	1.2	-2.4		
		MCDB	-5.1	-3.4	2.1	10.7	20.9	24.8	25.4	22.8	16.7	10.6	1.6	-2.0		
	10%	WB	-7.7	-6.4	-0.5	5.5	12.7	18.4	19.7	17.8	12.7	7.1	-0.2	-4.7		
		MCDB	-7.3	-5.6	1.0	9.0	18.2	23.2	23.9	20.8	14.8	8.7	0.2	-4.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	9.1	9.8	9.3	10.7	9.9	9.7	9.0	7.7	5.9	6.2	6.8		
		MCDBR	8.0	8.0	9.5	11.9	15.1	13.1	12.6	12.0	10.7	9.2	4.7	6.1		
	5% WB	MCWBR	7.8	7.2	6.8	7.5	7.5	6.1	5.7	6.4	6.4	6.0	4.5	5.8		
		MCWBR	7.8	7.5	8.0	11.2	13.9	11.7	11.6	11.1	10.0	8.8	4.7	6.3		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.308	0.365	0.395	0.398	0.405	0.416	0.420	0.365	0.336	0.290	0.259		
	taud		2.000	2.012	1.942	2.066	2.273	2.327	2.295	2.246	2.368	2.274	2.077	1.946		
	Ebn at Noon		479	661	731	793	828	828	806	767	756	644	449	334		
	Edh at Noon		51	87	128	136	119	115	116	114	85	68	47	36		
(44) All-Sky Solar Radiation	RadAvg		0.51	1.43	2.93	4.29	4.99	5.49	5.28	3.79	2.40	1.12	0.45	0.27		
	RadStd		0.04	0.11	0.19	0.46	0.41	0.40	0.50	0.30	0.24	0.14	0.08	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page